

Travel Booking Platform

Using Agile Methodology

Name: Lokesh Raghuwanshi

Branch: AI & DS

College: TIT, Bhopal

Semester: 7th

Subject: Software Engineering

1. Abstract (300–400 Words)

A Travel Booking Platform is an online system that allows users to search, book, and manage travel services such as flights, hotels, and tour packages. In today's fast-paced world, customers expect quick, reliable, and flexible booking solutions. This project focuses on developing a Travel Booking Platform using the **Agile Software Development Methodology**, which ensures continuous improvement, faster delivery, and customer satisfaction.

Traditional software development models often take a long time to deliver a complete system, and changes in user requirements are difficult to manage. Agile methodology overcomes these issues by dividing the project into small development cycles called **sprints**. Each sprint delivers a functional module, such as user registration, search functionality, booking management, or payment integration. Regular feedback from users helps improve the system continuously.

The platform allows users to create accounts, search for travel options, compare prices, make bookings, and manage their travel plans. Admin users can manage listings, monitor bookings, and handle customer queries. Agile practices like sprint planning, daily stand-ups, sprint reviews, and retrospectives help the development team stay organized and responsive to changes.

The system improves user experience by offering real-time availability, secure transactions, and easy navigation. It also benefits travel agencies by automating bookings, reducing manual work, and increasing operational efficiency.

This project demonstrates how Agile methodology helps in building scalable, user-friendly, and reliable software systems. It also provides hands-on experience with teamwork, iterative development, and real-world project management.

Future enhancements may include AI-based travel recommendations, mobile app integration, and cloud deployment to make the platform more advanced and accessible.

2. Introduction

2.1 Introduction

A Travel Booking Platform helps users plan and book travel services online. Agile methodology is used to ensure flexibility and faster development.

2.2 Problem Identification

Traditional booking systems are slow, rigid, and not user-friendly.

2.3 Need of the Project

A modern platform is required to provide easy, fast, and reliable travel bookings.

2.4 Project Scheduling

Phase	Duration
Planning	2 Days
Development	8 Days
Testing	3 Days
Review	2 Days
Documentation	1 Day

2.5 Objectives

- Provide online travel booking
- Improve user experience
- Use Agile methodology
- Ensure secure payments
- Enable real-time updates
- Enhance customer satisfaction

3. Software Requirement Specification (SRS)

3.1 Purpose

To develop a flexible and user-friendly Travel Booking Platform using Agile methodology.

3.2 Scope

Useful for travelers, travel agencies, and tourism businesses.

3.3 Hardware / Software Requirement

Hardware:

- 4GB RAM

- Intel i3 or above
- 500GB HDD

Software:

- Windows 10
- Web Technologies / Java
- MySQL
- VS Code / Eclipse

3.4 Tools

- HTML, CSS, JavaScript
- Java / Python
- MySQL
- Git
- Agile Tools (Jira / Trello)

3.5 Software Process Model

Agile Model

- Sprint Planning
- Daily Stand-up
- Sprint Review
- Sprint Retrospective

4. System Design

4.1 Data Dictionary

Field	Type	Description
UserID	int	Unique user ID
Name	String	User name
Destination	String	Travel location
BookingDate	Date	Date of booking
Status	String	Confirmed / Cancelled

4.2 ER Diagram

User → Booking → Travel Service

4.3 DFD

User → Platform → Database → Confirmation

4.4 Diagrams

- Use Case Diagram
- Activity Diagram
- Flowchart

5. Implementation

5.1 Program Code

Modules developed in sprints:

- User Registration & Login
- Search & Filter
- Booking Management
- Payment Integration
- Admin Dashboard

5.2 Output Screens

- Home Page
- Search Results
- Booking Page
- Payment Page
- User Dashboard

6. Testing

6.1 Test Data

Input	Expected Output
Invalid Card	Payment Failed
Empty Fields	Validation Error
Duplicate Booking Warning	

6.2 Test Result

All features worked successfully after sprint-based testing.

7. User Manual

7.1 How to Use

1. Register/Login
2. Search travel options
3. Select and book
4. Make payment
5. View bookings

7.2 Screen Layout

Web-based interface with simple navigation.

8. Project Applications & Limitations

Applications

- Travel Agencies
- Online Tourism
- Corporate Travel
- Vacation Planning

Limitations

- Internet dependency
- Limited offline support
- Manual updates required

9. Conclusion & Future Enhancement

The Travel Booking Platform developed using Agile methodology provides a flexible and efficient solution for online travel planning. Agile ensures faster delivery, better quality, and continuous improvement.

Future Enhancements:

- AI travel suggestions
- Mobile app version

- Cloud hosting
- Chatbot support

10. Bibliography & References

- Agile Manifesto
- Scrum Guide
- Tourism Websites
- GeeksForGeeks
- IEEE Journals